Java Lab Problem: Warehouse Robot Inventory System

Topics: OOP + 2D Arrays + Spiral Traversal

Problem Statement

A warehouse stores products in a 2D grid, where each cell represents the number of products stored in a particular section. Create a Java class **Warehouse** to manage this inventory. The warehouse is represented by an N × N 2D array.

Requirements

1. Create a class **Warehouse** with a 2D array named **inventory**.

2. Use a **constructor** to initialize the warehouse inventory.

3. Create **displayInventory()** to display the warehouse layout.

4. The warehouse robot starts from the top-left corner and visits every section by moving around the boundary and then moving inward.

5. Create **getRobotVisitOrder()** to print the number of products in each section in **spiral traversal order**.

6. Create **calculateBoundaryStock()** to calculate the total products stored in all boundary sections.

7. Create **findMaximumSection()** to identify the section containing the maximum number of products.

8. Create an object of **Warehouse** in **main()** and demonstrate all operations.

Example Input

```
10  20  30  40
50  60  70  80
90 100 110 120
130 140 150 160
```

Robot Visit Order

```
10 20 30 40 80 120 160 150
140 130 90 50 60 70 110 100
```

Expected Information

The program should display the total boundary stock and identify the section containing the maximum stock value (160 in the example).

OOP Concepts Tested

Class, Object, Constructor, Encapsulation, Methods.

2D Array Concepts Tested

Nested loops, row/column traversal, boundary traversal, spiral traversal, and finding the maximum value.